

3. Project Design Phase

ComicCraft – AI Comic Story Creator Using Gemini Models

Designing the application architecture, modules, and user flow

1. Design Objective

The design organizes ComicCraft into cooperating frontend, backend, and AI-integration components. Each component has a defined responsibility in the end-to-end comic-generation process.

2. High-Level Architecture

Frontend: HTML, CSS, and Jinja2 templates collect user input and display results. Backend: FastAPI receives requests, coordinates processing, handles routes, and prepares output. AI integration: Gemini APIs generate story content and Stable Diffusion generates illustrations.

3. Frontend Design

The homepage is index.html and contains the story prompt, main character name, setting, tone, and art-style fields. comic_preview.html presents the generated panels sequentially. export_success.html confirms that the comic has been exported. The guide describes a scenic background, centered form, clear typography, rounded elements, shadows, and button hover effects.

4. Backend Design

The main routing logic is placed in routes.py. The backend accepts form data or JSON, calls the generation modules, builds the comic layout, and returns a rendered page or JSON response. FastAPI HTTPException and try-except blocks are used for error handling.

5. AI Module Design

gemini_flash.py contains generate_outline() for the structured five-panel outline. gemini_pro.py contains generate_story() for narration and dialogue. image_generator.py contains generate_image() for panel illustrations and saves images to static/panels.

6. Layout and Export Design

layout_builder.py provides build_comic_layout() to pair each image with its panel story. exporters.py provides save_pdf() to create a multi-page PDF with panel images and narration, saved in static/exports.

7. Route Design

The documented routes are / for the homepage, /generate for form-based comic generation, /generate-comic/json for JSON-based generation, /export-success for export confirmation, and /test-image for direct image-generation testing.

8. User Flow

The user opens the homepage, enters preferences, submits the form, waits while the backend generates the outline, story, and images, reviews the preview, and downloads the PDF. The application then displays an export-success page.

9. Design Outcome

The design phase defines the application's component boundaries, module responsibilities, templates, routes, and user journey. These design decisions guide the development and integration work.